

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458109.7699 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1055 +/- 0.0088
Transit depth (Rp/Rs)^2	1.11 +/- 0.18 [%]
Orbital Inclination (inc)	86.03 +/- 0.52
Airmass coefficient 1 (a1)	2.177 +/- 0.015
Airmass coefficient 2 (a2)	-0.572 +/- 0.026
Scatter in the residuals of the lightcurve fit is	0.67 %
Best Comparison Star	None
Optimal Aperture	5.3
Optimal Annulus	10.21
Transit Duration (day)	0.1157 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1055 ± 0.0088 vs 0.11 (deviation 0.32σ)
Duration fit vs published	0.1157 d vs 0.1208 d (deviation 0.53σ)
Mid-time O–C	5.8 min
Depth SNR	7.5
Residual scatter	0.67 %

Light curve

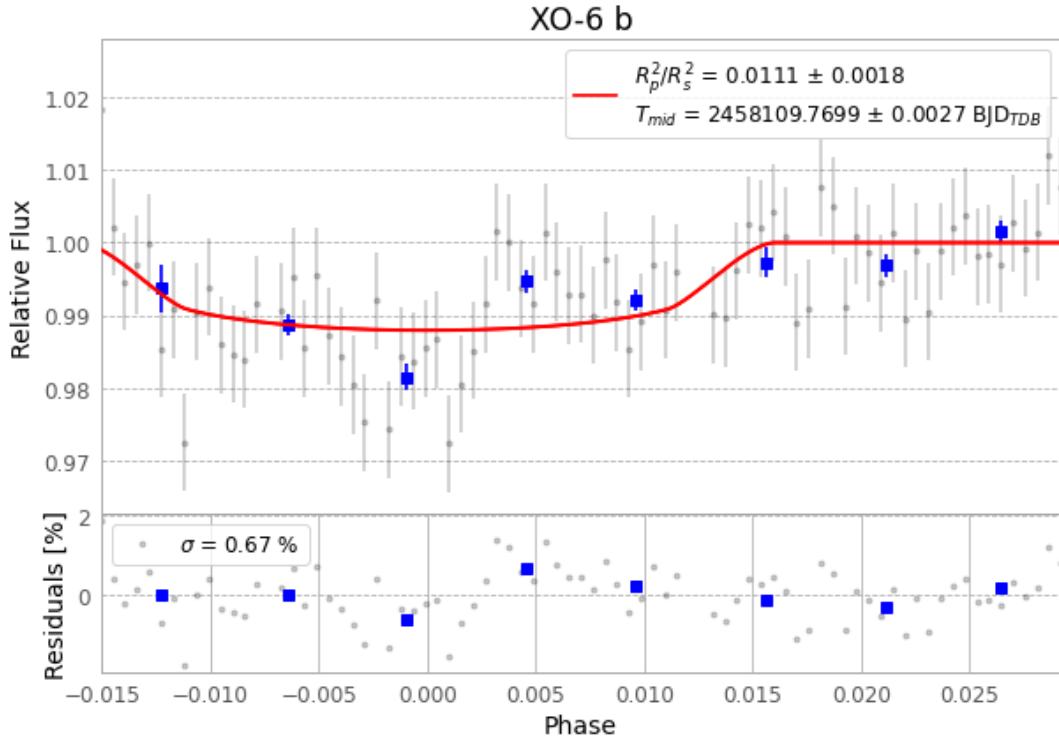


Figure 1 — FinalLightCurve_XO-6 b_2017-12-21.png (SHA-256 9f42b8bb97531cfd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [527, 354], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 16637a73011c130d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

79 FITS frames (52 MB), XO-6171222050012.FITS → XO-6171222090012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-6-171222.log (57d29e529555...), FinalLightCurve_XO-6 b_2017-12-21.png (9f42b8bb9753...), FinalLightCurve_XO-6 b_2017-12-21.csv (71083ab8d3ce...)

Compute resources

Wall time 195.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 23:43Z.